

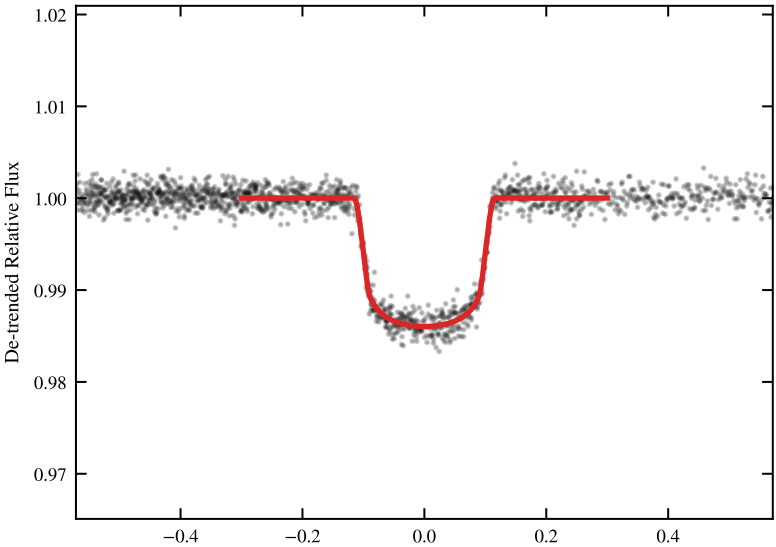
TESS PLANET TRANSIT CHARACTERIZATION REPORT

Target: TIC 207081058 | RA: 331.86752° | Dec: -41.81551° | Sectors: 105, 106

Derived Planetary Parameters

Parameter	Derived Value (MCMC)
Period (d)	14.775 ± 0.000263
Rp/R★	0.11004 ± 0.00103
Rp (R⊕)	14.75 ± 0.591
b	0.19909 ± 0.103
T14 (h)	5.4875 ± 0.0333
a (AU)	0.12772 ± 0.00552
Teq (K)	907.76 ± 27.3

Primary Phased Transit



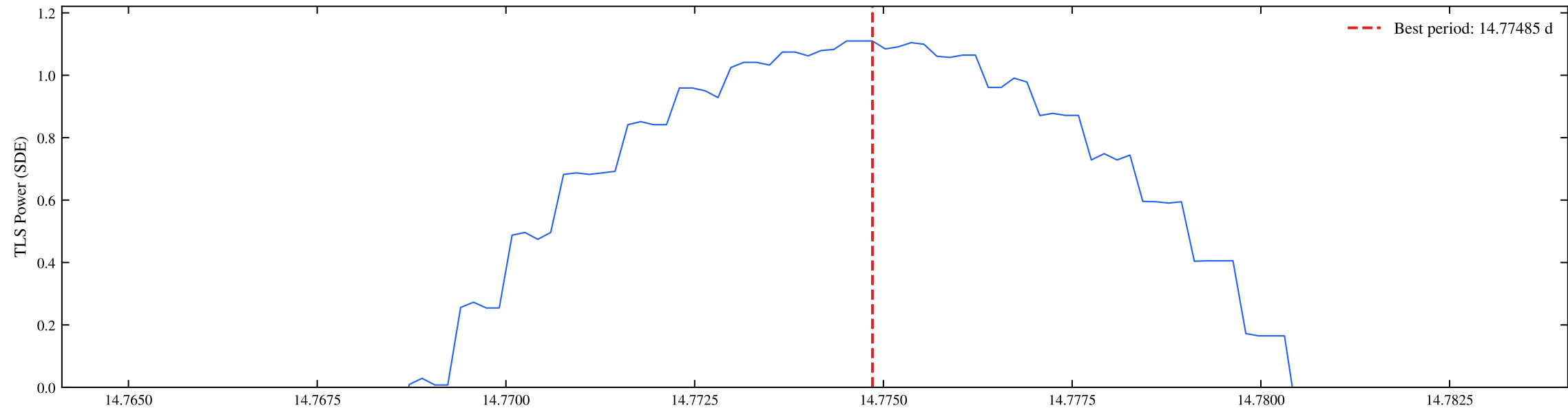
Literature Comparison & MCMC Diagnostics

Parameter / Metric	Literature Value	Derived / Diagnostic
P1 Period (d)	14.7749	14.774896 ± 0.00026
P1 Epoch (BTJD)	N/A	4217.2657 ± 0.0004
Max R-hat (R)	N/A	1.0016
Min ESS	N/A	3668
MCMC Divergences	N/A	1

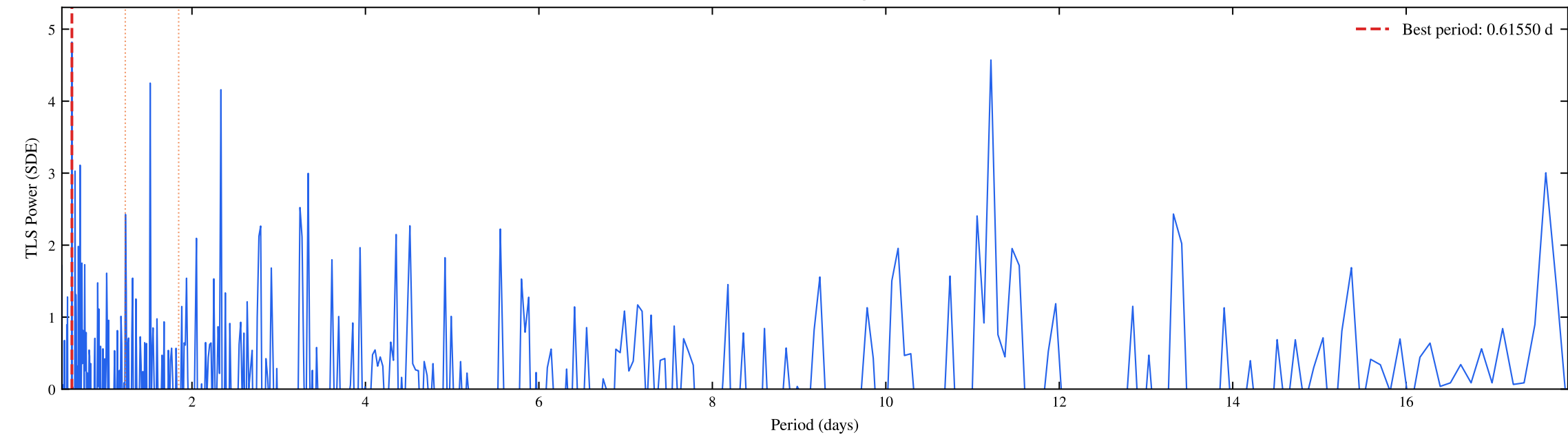
Host Star Properties

Stellar Parameter	Value	Source
R★ (R☉)	1.227 ± 0.048	VizieR TIC8
M★ (M☉)	1.14 ± 0.123	VizieR TIC8
Teff (K)	6073 ± 173	VizieR TIC8
log g (dex)	4.317 ± 0.0912	VizieR TIC8
[Fe/H]	-0.15	SIMBAD
ρ★ (g/cm³)	0.87 ± 0.139	Derived

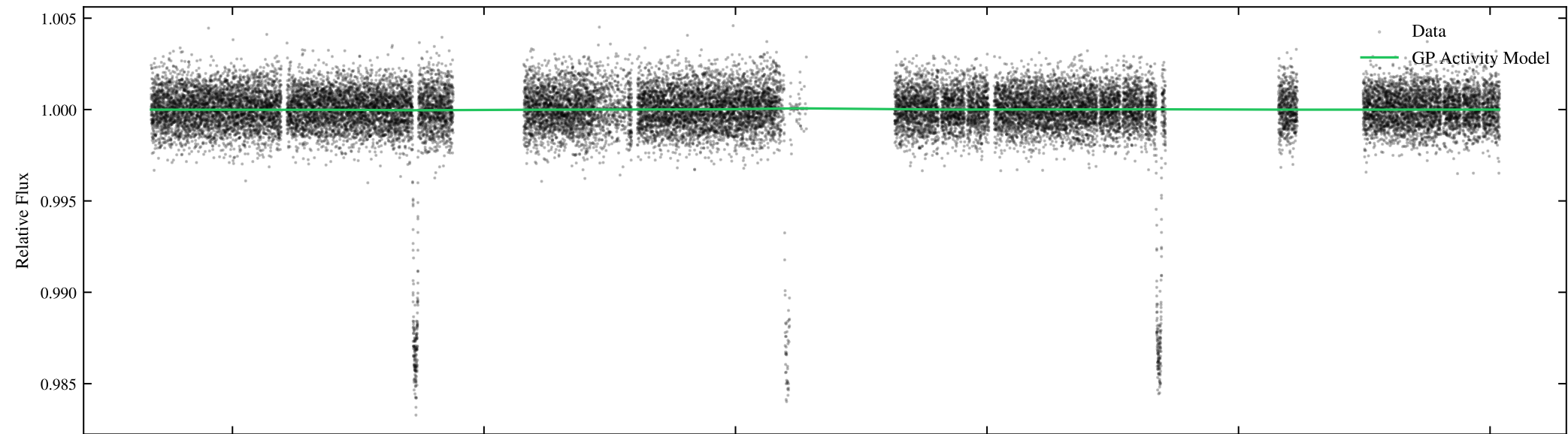
TIC 207081058 | Sectors: 105, 106 | TLS Periodogram (Planet 1 Search) (SDE = 1.11)



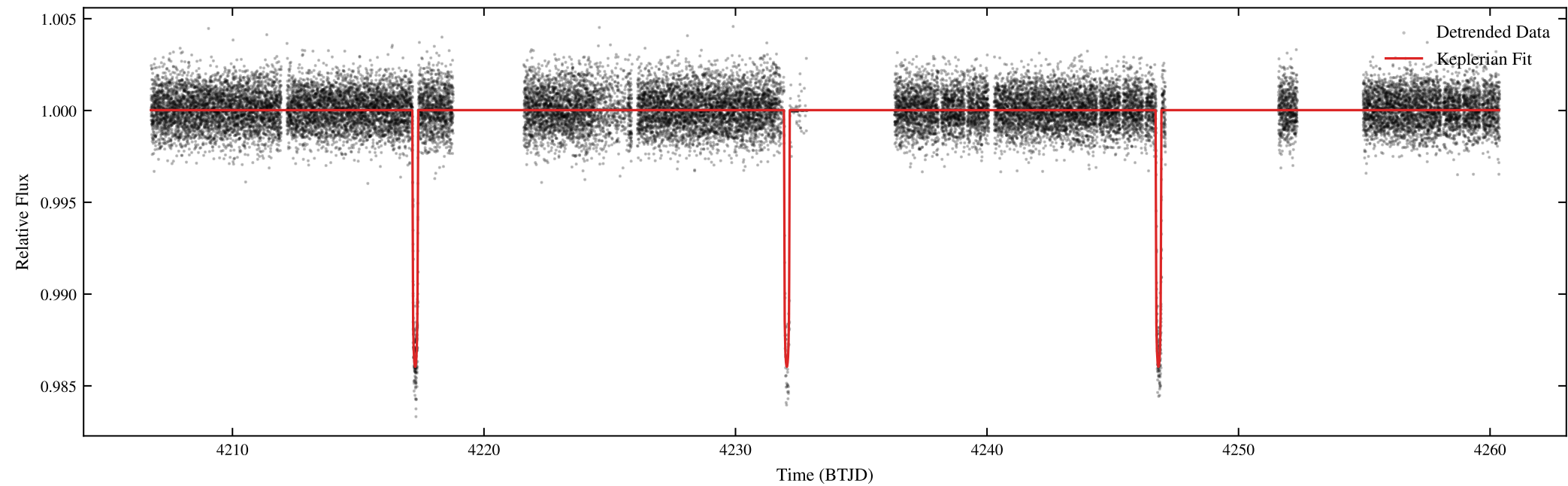
TIC 207081058 | Sectors: 105, 106 | TLS Periodogram (Planet 2 Search) (SDE = 4.82)



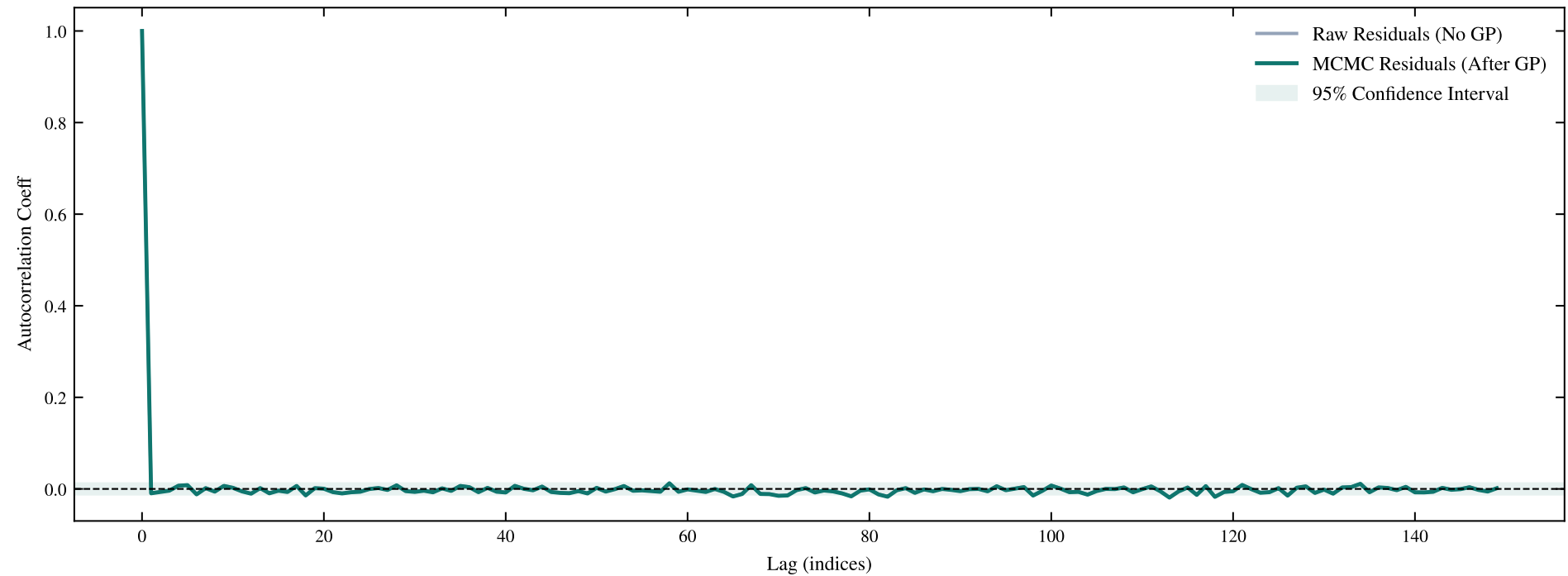
TIC 207081058 | Sectors: 105, 106 | Bayesian GP Detrending (Time Domain)



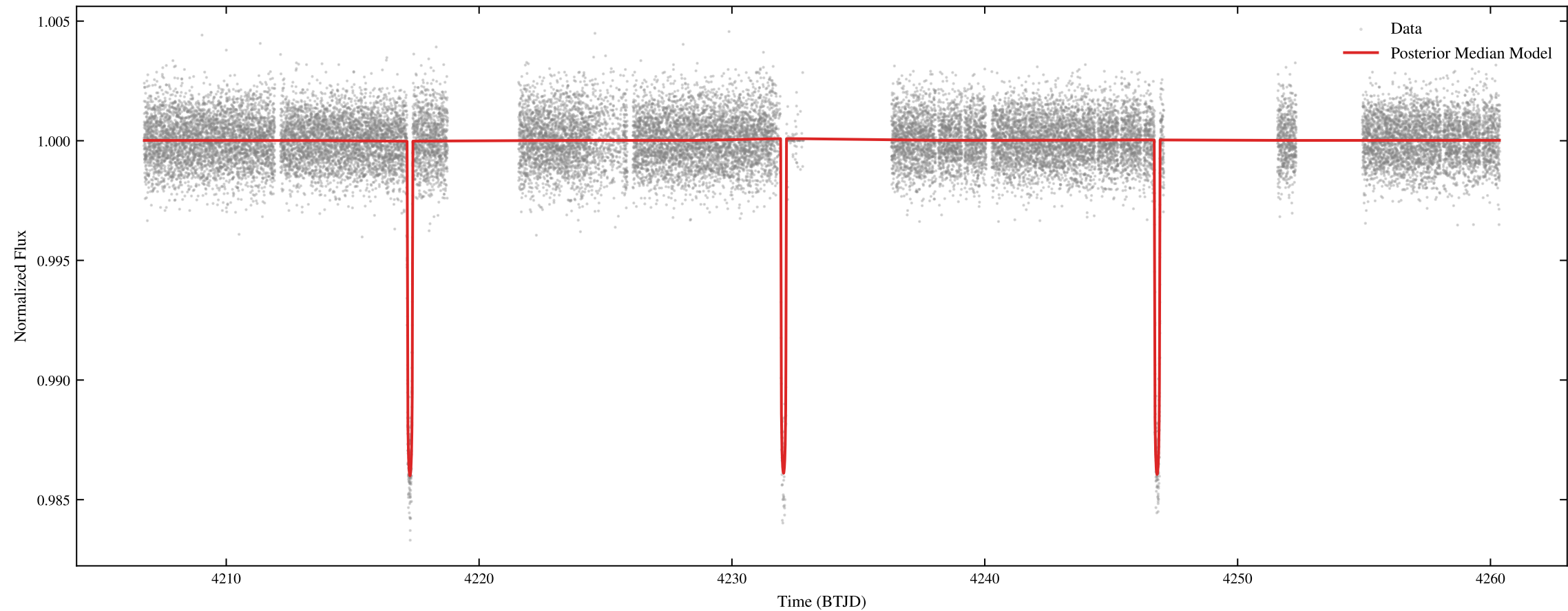
Detrended Light Curve & Best-Fit Transit Model

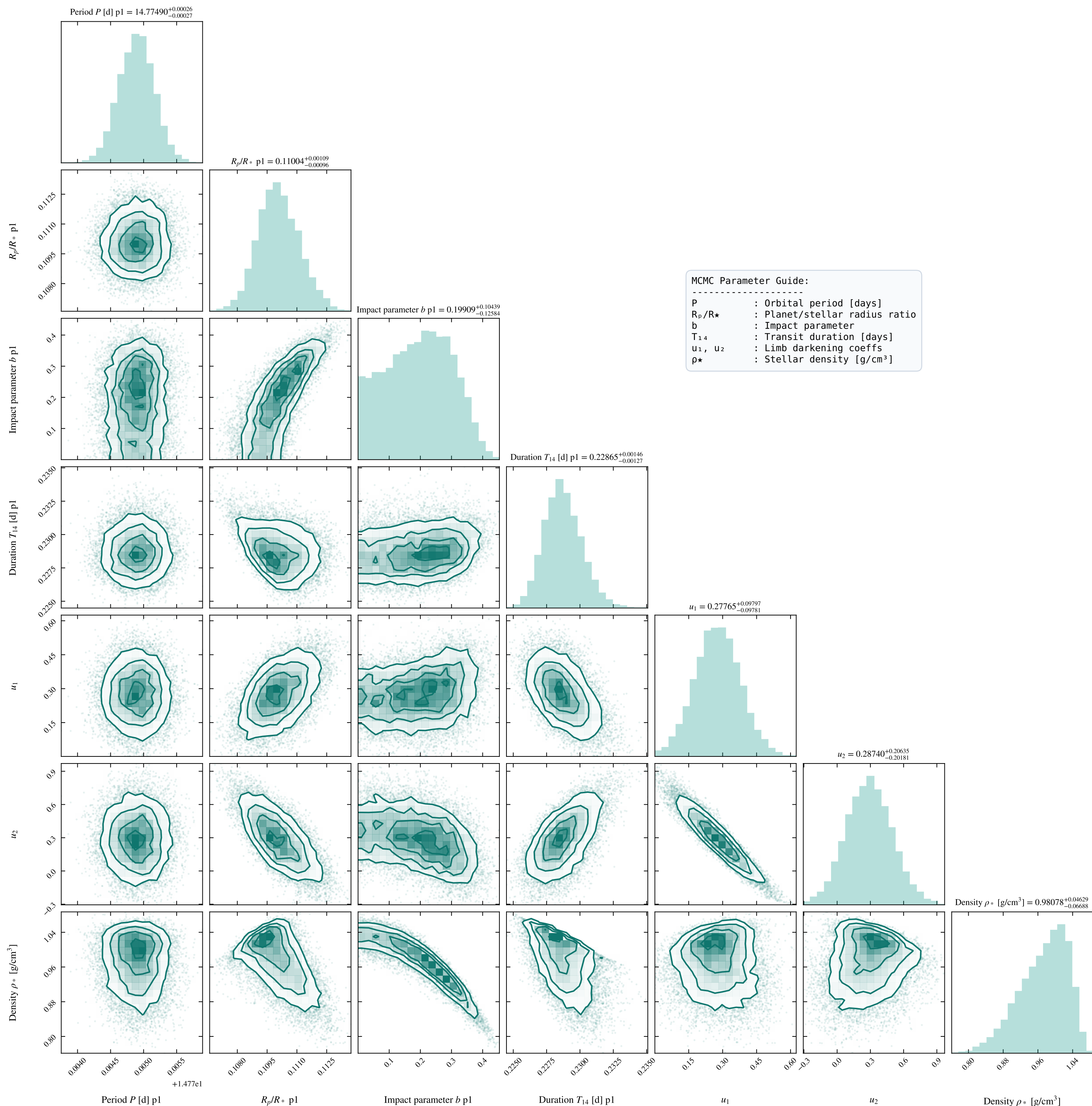


TIC 207081058 | Sectors: 105, 106 | GP Residuals Autocorrelation Function (ACF)

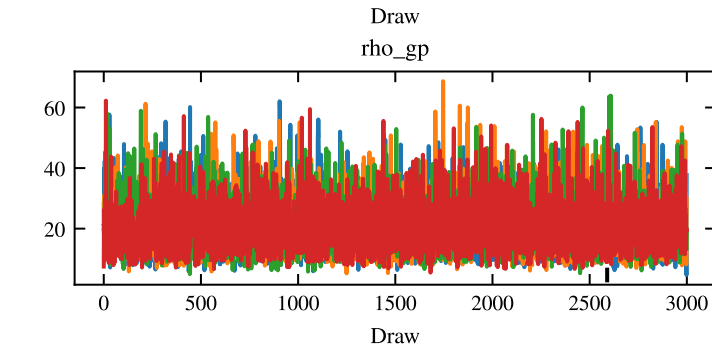
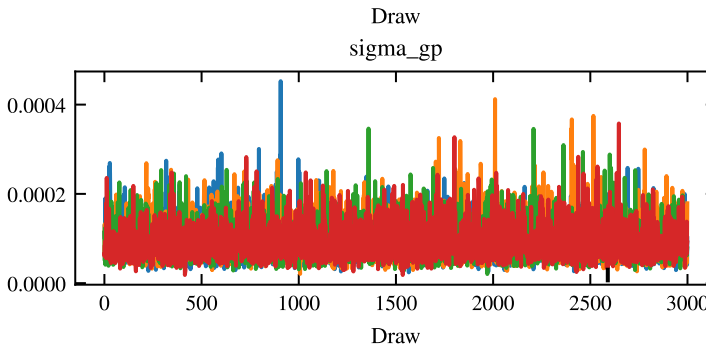
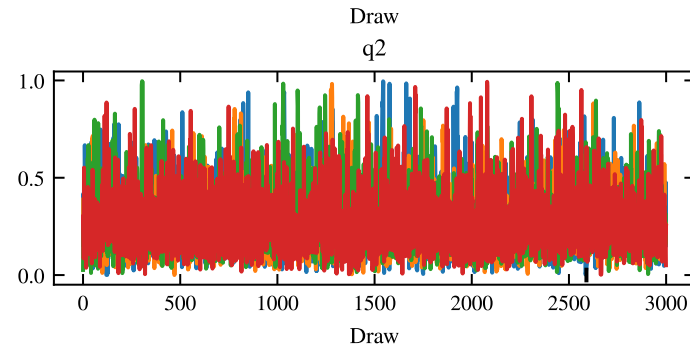
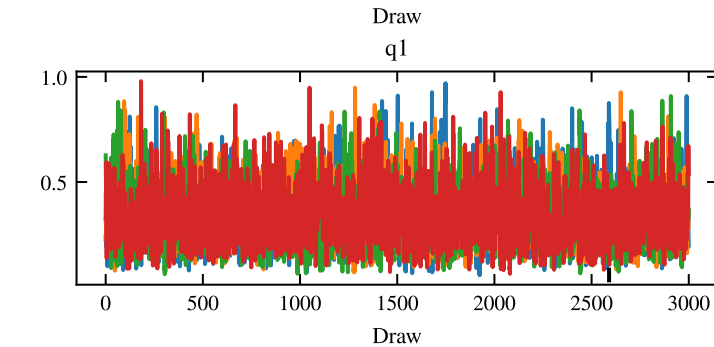
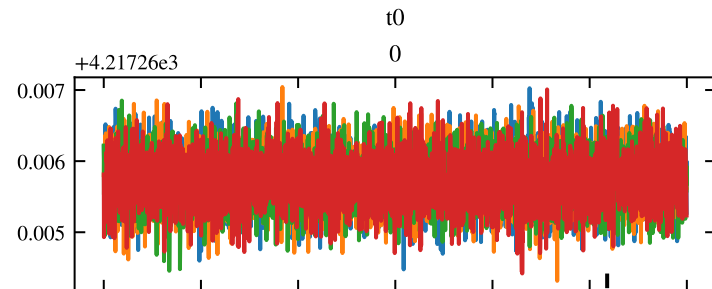
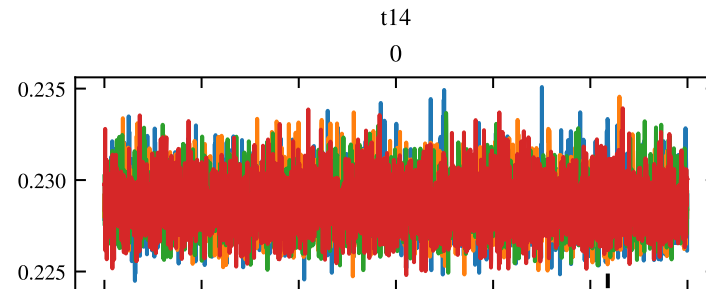
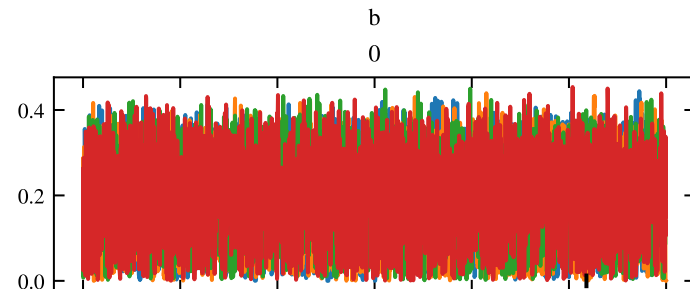
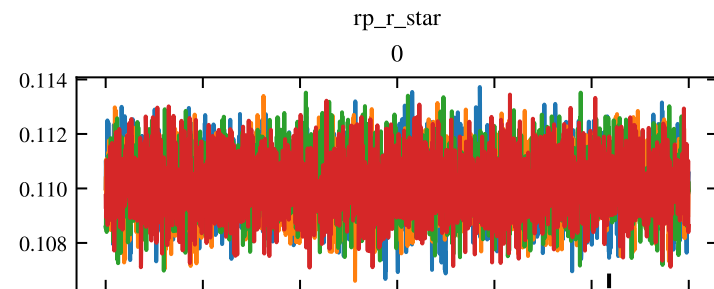


TIC 207081058 | Sectors: 105, 106 | MCMC Posterior Predictive Check

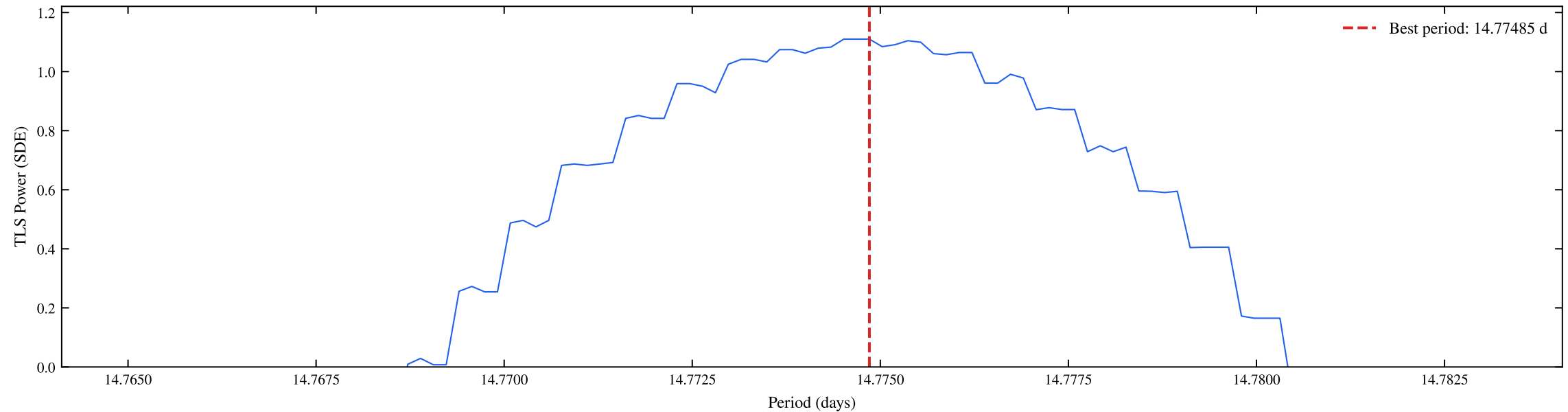




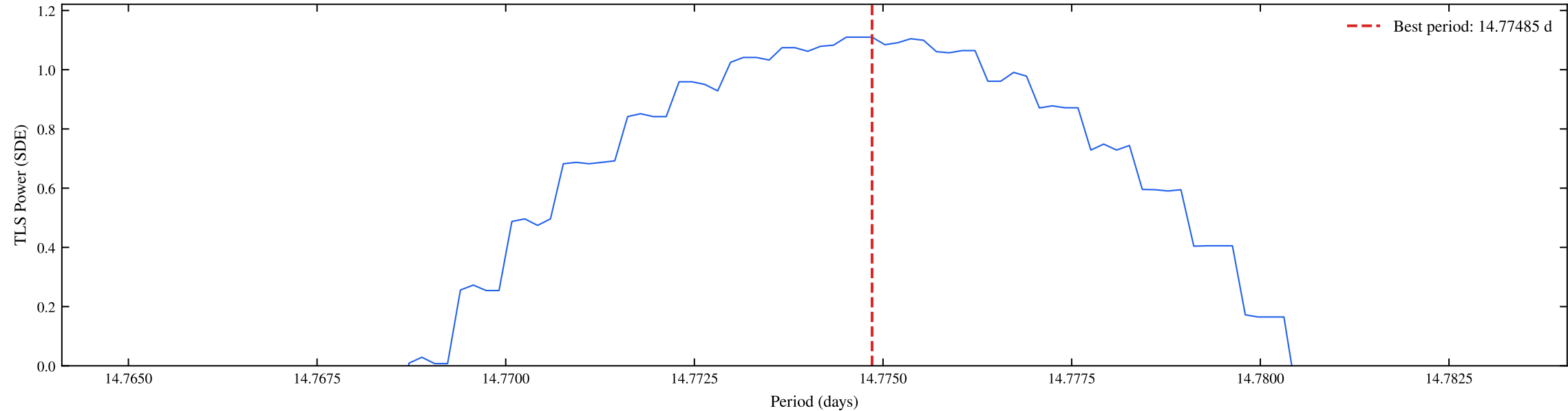
TIC 207081058 | Sectors: 105, 106 | MCMC Trace Plots



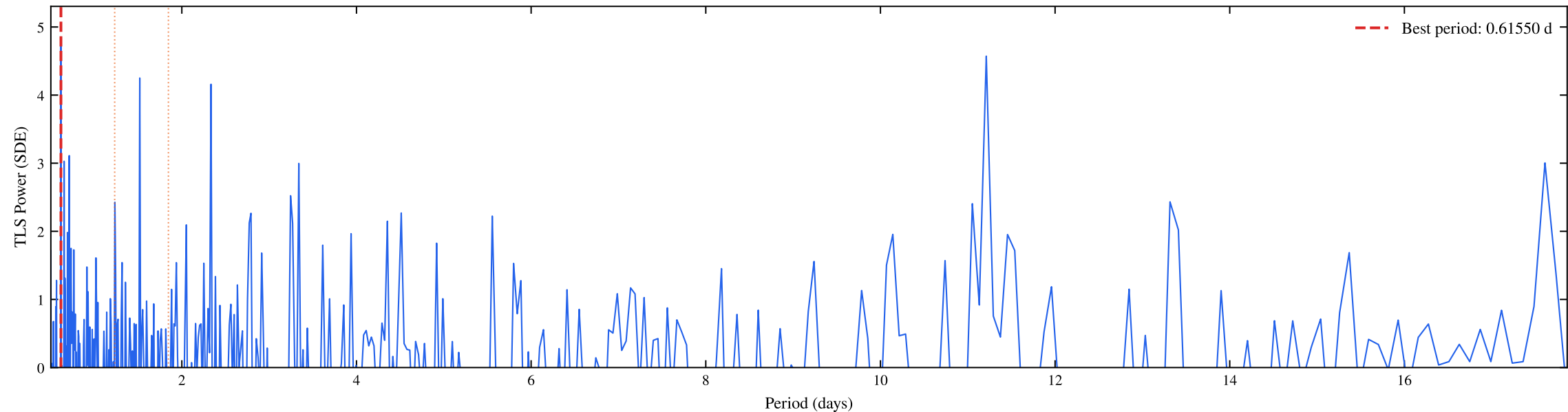
TIC 207081058 | Sectors: 105, 106 | TLS Coarse Periodogram (SDE = 1.11)



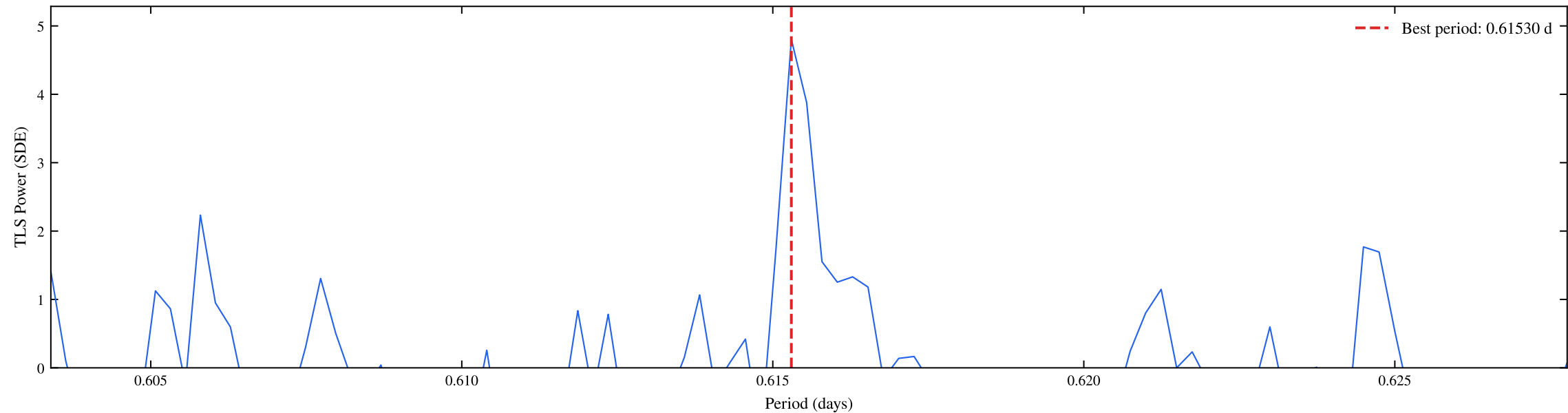
TIC 207081058 | Sectors: 105, 106 | TLS Fine Periodogram (SDE = 1.11)



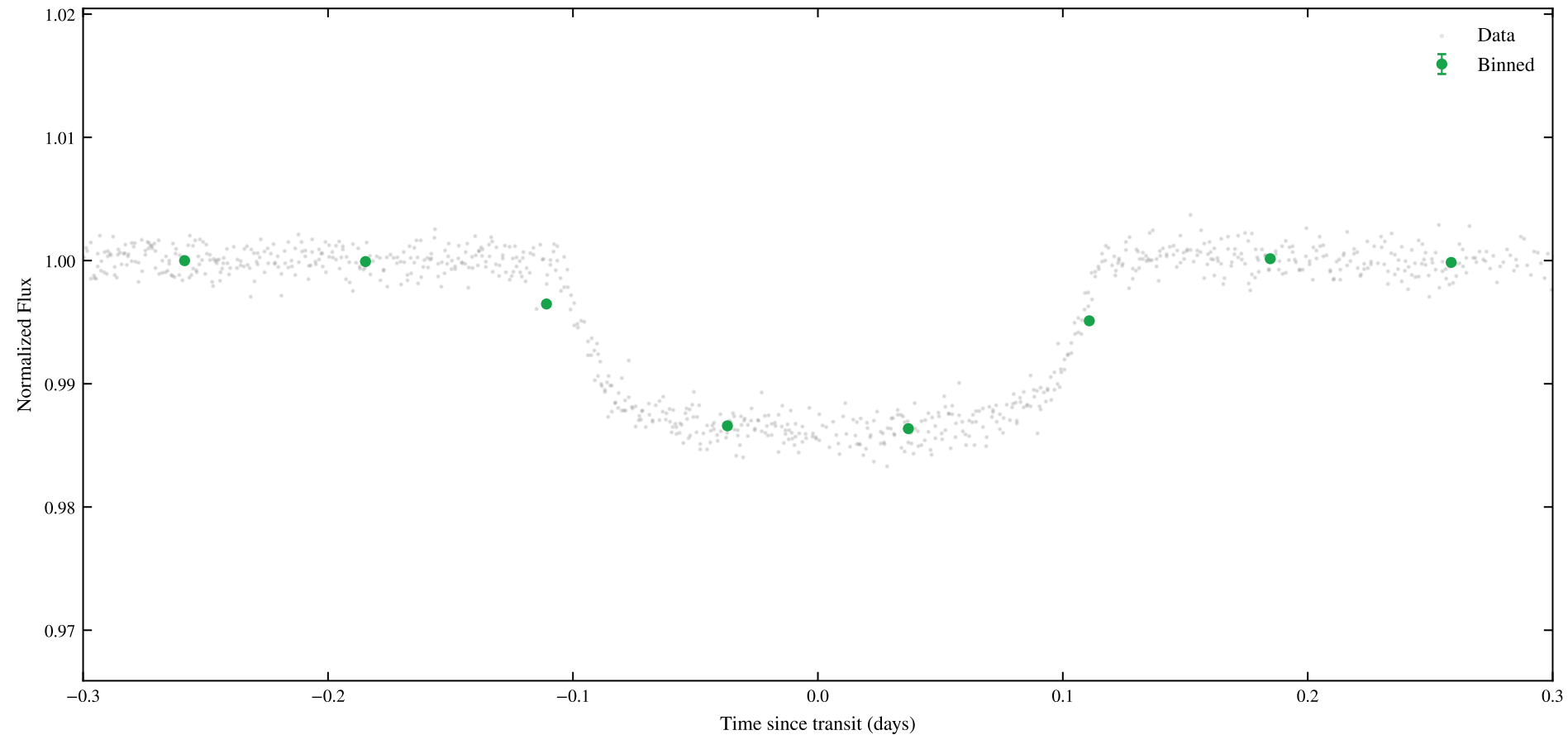
TIC 207081058 | Sectors: 105, 106 | TLS Coarse Periodogram (SDE = 4.82)



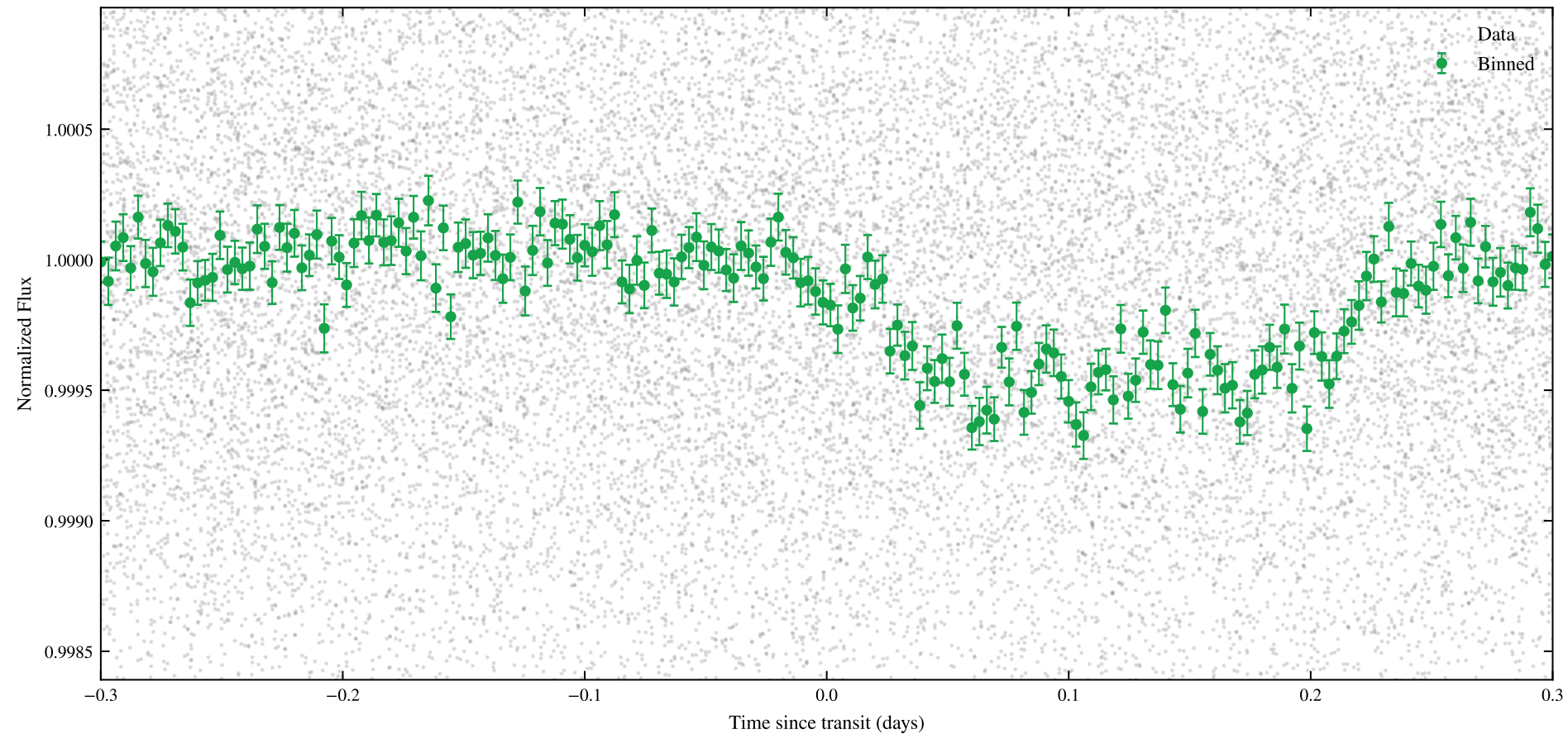
TIC 207081058 | Sectors: 105, 106 | TLS Fine Periodogram (SDE = 4.81)



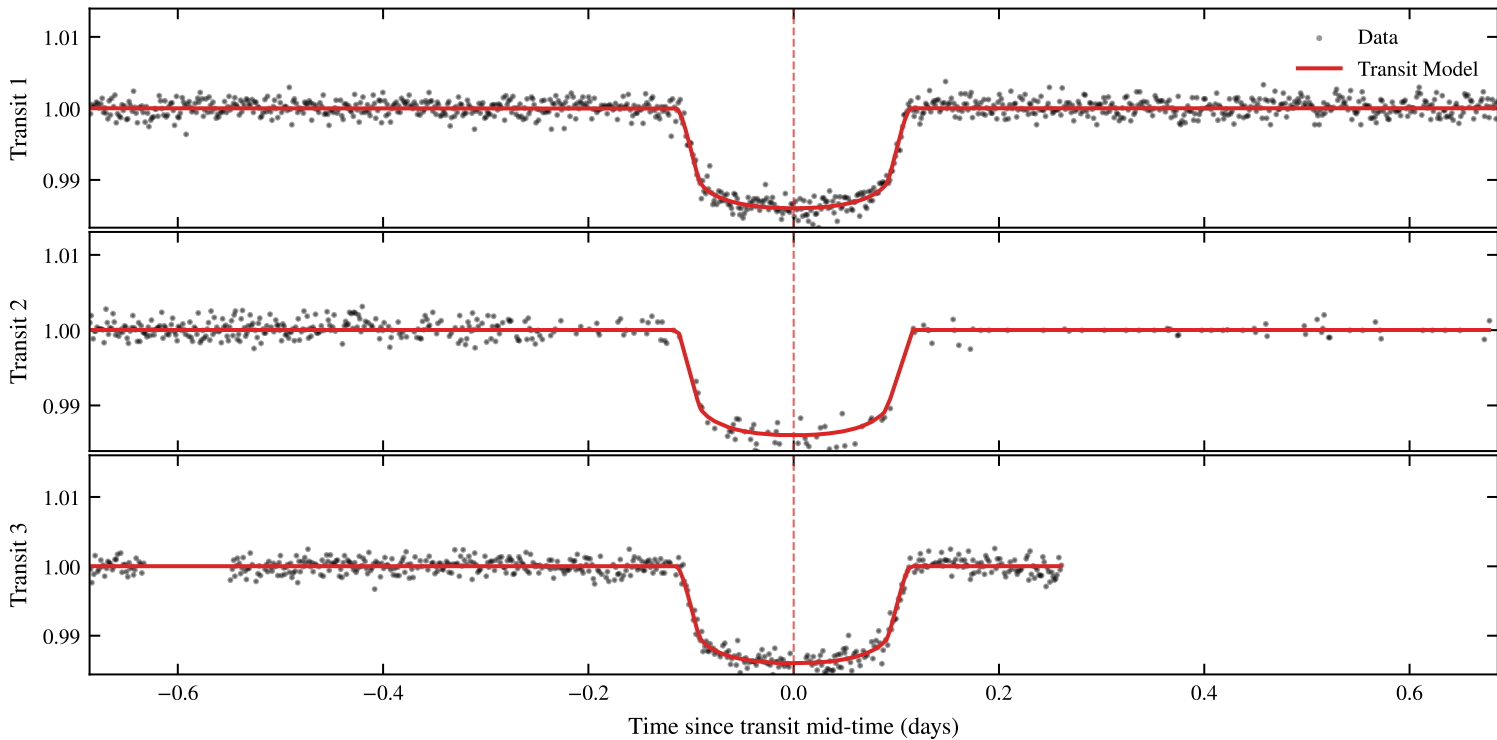
TIC 207081058 | Sectors: 105, 106 | Phase-Folded Transit (P = 14.77485 d)



TIC 207081058 | Sectors: 105, 106 | Phase-Folded Transit (P = 0.61530 d)



TIC 207081058 | Sectors: 105, 106 | Stacked Individual Transits



TIC 207081058 | Sectors: 105, 106 | Phased MCMC Fit - Planet 1

